

Name: Daniel Oko Tettey

ID: 44352028

Cohort: B

CREATE DATABASE Projects;

USE Projects;

```
CREATE TABLE Employee (  
    empNo VARCHAR(5) PRIMARY KEY,  
    fName VARCHAR(20),  
    lName VARCHAR(20),  
    address VARCHAR(30),  
    DOB DATE,  
    sex VARCHAR(10),  
    position VARCHAR(15),  
    deptNo VARCHAR(5),  
  
    CONSTRAINT chk_sex CHECK(sex IN ("M", "F")),  
    CONSTRAINT chk_position CHECK(position IN ("Manager", "Team  
Leader", "Analyst", "Software Developer"))  
);
```

```
CREATE TABLE Department (  
    deptNo VARCHAR(5) PRIMARY KEY,  
    deptName VARCHAR(20),  
    mgrEmpNo VARCHAR (5)  
);
```

```
CREATE TABLE Project (  
    projNo VARCHAR(5) PRIMARY KEY,  
    projName VARCHAR(20),  
    deptNo VARCHAR(5),  
  
    CONSTRAINT fk_deptNo  
    FOREIGN KEY (deptNo)  
    REFERENCES Department(deptNo)  
);
```

```
CREATE TABLE WorksOn (  
    empNo VARCHAR(5),  
    projNo VARCHAR(5),  
    dateWorked DATE,  
    hoursWorked INT(3),  
  
    CONSTRAINT pkWorksOn  
    PRIMARY KEY(empNo, projNo, dateWorked),
```

```

CONSTRAINT fk_empNo
FOREIGN KEY (empNo)
REFERENCES Employee(empNo),

CONSTRAINT fk_projNo
FOREIGN KEY (projNo)
REFERENCES Project(projNo)
);

ALTER TABLE Employee
ADD CONSTRAINT fk_EmpDept
FOREIGN KEY (deptNo)
REFERENCES Department(deptNo);

ALTER TABLE WorksOn
ADD CONSTRAINT chk_hoursW CHECK(hoursWorked >0 AND hoursWorked <=
40);

ALTER TABLE Department
ADD CONSTRAINT fk_mgEmpNo
FOREIGN KEY (mgrEmpNo)
REFERENCES Employee(empNo);

SET FOREIGN_KEY_CHECKS = 0;

INSERT INTO Department (deptNo, deptName, mgrEmpNo) VALUES
('D0001', 'Frontend Dev', 'E0001'),
('D0002', 'Backend Dev', 'E0002'),
('D0003', 'Database Admin', 'E0003'),
('D0004', 'QA Testing', 'E0001'),
('D0005', 'UI/UX Design', 'E0002'),
('D0006', 'Cloud Ops', 'E0003'),
('D0007', 'Data Science', 'E0001'),
('D0008', 'Cybersecurity', 'E0002'),
('D0009', 'Tech Support', 'E0003'),
('D0010', 'R&D', 'E0001');

SELECT * FROM Department;

+-----+-----+-----+
| deptNo | deptName    | mgrEmpNo |
+-----+-----+-----+
| D0001 | Frontend Dev | E0001    |
| D0002 | Backend Dev  | E0002    |
| D0003 | Database Admin | E0003    |
| D0004 | QA Testing   | E0001    |
| D0005 | UI/UX Design | E0002    |

```

```

| D0006 | Cloud Ops    | E0003 |
| D0007 | Data Science  | E0001 |
| D0008 | Cybersecurity | E0002 |
| D0009 | Tech Support  | E0003 |
| D0010 | R&D           | E0001 |
+-----+-----+-----+

```

```

INSERT INTO Employee (empNo, fName, lName, address, DOB, sex, position,
deptNo) VALUES
('E0001', 'Alice', 'Smith', '123 Maple St', '1980-01-10', 'F', 'Manager', 'D0001'),
('E0002', 'Bob', 'Jones', '456 Oak St', '1982-02-20', 'M', 'Manager', 'D0002'),
('E0003', 'Charlie', 'Brown', '789 Pine St', '1985-03-30', 'M', 'Manager',
'D0003'),
('E0004', 'Diana', 'Prince', '111 Elm St', '1990-04-15', 'F', 'Leader', 'D0001'),
('E0005', 'Evan', 'Wright', '222 Cedar St', '1992-05-25', 'M', 'Leader', 'D0002'),
('E0006', 'Fiona', 'Gallagher', '333 Birch St', '1994-06-05', 'F', 'Analyst',
'D0003'),
('E0007', 'George', 'Clark', '444 Ash St', '1995-07-12', 'M', 'Analyst', 'D0004'),
('E0008', 'Hannah', 'Abbott', '555 Walnut St', '1996-08-22', 'F', 'Developer',
'D0001'),
('E0009', 'Ian', 'Malcolm', '666 Cherry St', '1997-09-09', 'M', 'Developer',
'D0002'),
('E0010', 'Julia', 'Stiles', '777 Spruce St', '1998-10-31', 'F', 'Developer', 'D0003');

SELECT * FROM Employee;

```

```

+-----+-----+-----+-----+-----+-----+-----+
+-----+
| empNo | fName | lName | address | DOB | sex | position | deptNo |
+-----+-----+-----+-----+-----+-----+-----+
+-----+
| E0001 | Alice | Smith | 123 Maple St | 1980-01-10 | F | Manager | D0001 |
| E0002 | Bob | Jones | 456 Oak St | 1982-02-20 | M | Manager | D0002 |
|
| E0003 | Charlie | Brown | 789 Pine St | 1985-03-30 | M | Manager |
D0003 |
| E0004 | Diana | Prince | 111 Elm St | 1990-04-15 | F | Leader | D0001 |
| E0005 | Evan | Wright | 222 Cedar St | 1992-05-25 | M | Leader | D0002 |
|
| E0006 | Fiona | Gallagher | 333 Birch St | 1994-06-05 | F | Analyst | D0003 |
|
| E0007 | George | Clark | 444 Ash St | 1995-07-12 | M | Analyst | D0004 |
|
| E0008 | Hannah | Abbott | 555 Walnut St | 1996-08-22 | F | Developer |
D0001 |
| E0009 | Ian | Malcolm | 666 Cherry St | 1997-09-09 | M | Developer |

```

```

D0002 |
| E0010 | Julia | Stiles | 777 Spruce St | 1998-10-31 | F | Developer | D0003 |
+-----+-----+-----+-----+-----+-----+-----+
+-----+

```

```

INSERT INTO Project (projNo, projName, deptNo) VALUES
('P0001', 'Web Redesign', 'D0001'),
('P0002', 'API Migration', 'D0002'),
('P0003', 'DB Optimization', 'D0003'),
('P0004', 'Automated Tests', 'D0004'),
('P0005', 'Mobile App UI', 'D0005'),
('P0006', 'AWS Server Move', 'D0006'),
('P0007', 'AI Chatbot', 'D0007'),
('P0008', 'Security Audit', 'D0008'),
('P0009', 'Helpdesk Portal', 'D0009'),
('P0010', 'VR Prototype', 'D0010');

```

```

SELECT * FROM Project;

```

```

+-----+-----+-----+
| projNo | projName | deptNo |
+-----+-----+-----+
| P0001 | Web Rede... | D0001 |
| P0002 | API Migr... | D0002 |
| P0003 | DB Optim... | D0003 |
| P0004 | Automate... | D0004 |
| P0005 | Mobile A... | D0005 |
| P0006 | AWS Serv... | D0006 |
| P0007 | AI Chatb... | D0007 |
| P0008 | Security... | D0008 |
| P0009 | Helpdesk... | D0009 |
| P0010 | VR Proto... | D0010 |
+-----+-----+-----+

```

```

INSERT INTO WorksOn (empNo, projNo, dateWorked, hoursWorked) VALUES
('E0008', 'P0001', '2026-06-01', 30),
('E0004', 'P0001', '2026-06-01', 40),
('E0009', 'P0002', '2026-06-02', 35),
('E0005', 'P0002', '2026-06-02', 40),
('E0010', 'P0003', '2026-06-03', 20),
('E0006', 'P0003', '2026-06-03', 25),
('E0007', 'P0004', '2026-06-04', 38),
('E0001', 'P0007', '2026-06-05', 10),

```

```
('E0002', 'P0008', '2026-06-06', 15),
('E0003', 'P0006', '2026-06-07', 20);
```

```
SELECT * FROM WorksOn;
```

```
+-----+-----+-----+-----+
| empNo | projNo | dateWorked | hoursWorked |
+-----+-----+-----+-----+
| E0001 | P0007 | 2026-06-05 | 10          |
| E0002 | P0008 | 2026-06-06 | 15          |
| E0003 | P0006 | 2026-06-07 | 20          |
| E0004 | P0001 | 2026-06-01 | 40          |
| E0005 | P0002 | 2026-06-02 | 40          |
| E0006 | P0003 | 2026-06-03 | 25          |
| E0007 | P0004 | 2026-06-04 | 38          |
| E0008 | P0001 | 2026-06-01 | 30          |
| E0009 | P0002 | 2026-06-02 | 35          |
| E0010 | P0003 | 2026-06-03 | 20          |
+-----+-----+-----+-----+
```

```
SET FOREIGN_KEY_CHECKS = 1;
```

```
ALTER TABLE Employee
ADD CONSTRAINT chk_position CHECK (position IN ("Manager", "Leader",
"Developer", "Analyst"));
```

```
ALTER TABLE Employee
DROP CHECK chk_position;
```

```
ALTER TABLE Employee
MODIFY position VARCHAR(25);
```

```
UPDATE Employee
SET position = "Team Leader"
WHERE position = "Leader";
```

```
UPDATE Employee
SET position = "Software Developer"
WHERE position = "Developer";
```

```
ALTER TABLE Employee
ADD CONSTRAINT chk_position CHECK(position IN("Manager", "Team Leader",
"Software Developer", "Analyst"));
```

```
CREATE VIEW pManagedByF AS
```

```

SELECT p.projNo, p.projName
FROM Project p
JOIN Department d ON p.deptNo = d.deptNo
JOIN Employee e ON e.empNo = d.mgrEmpNo
WHERE e.sex = 'F' AND e.position = 'Manager'
ORDER BY p.projNo;

```

```

SELECT * FROM pManagedByF;

```

```

+-----+-----+
| projNo | projName      |
+-----+-----+
| P0001 | Web Redesign  |
| P0004 | Automated Tests |
| P0007 | AI Chatbot    |
| P0010 | VR Prototype  |
+-----+-----+

```

```

CREATE VIEW custAtributes AS
SELECT e.empNo, e.fname, e.lName, p.projName, w.hoursWorked
FROM Employee e, Project p, WorksOn w
WHERE e.empNo = w.empNo AND p.projNo = w.projNo;

```

```

SELECT * FROM custAtributes;

```

```

+-----+-----+-----+-----+-----+
| empNo | fname  | lName  | projName      | hoursWorked |
+-----+-----+-----+-----+-----+
| E0001 | Alice  | Smith  | AI Chatbot    | 10          |
| E0002 | Bob    | Jones  | Security Audit | 15          |
| E0003 | Charlie | Brown  | AWS Server Move | 20          |
| E0004 | Diana  | Prince | Web Redesign  | 40          |
| E0005 | Evan   | Wright | API Migration | 40          |
| E0006 | Fiona  | Gallagher | DB Optimization | 25          |
| E0007 | George | Clark  | Automated Tests | 38          |
| E0008 | Hannah | Abbott | Web Redesign  | 30          |
| E0009 | Ian    | Malcolm | API Migration | 35          |
| E0010 | Julia  | Stiles  | DB Optimization | 20          |
+-----+-----+-----+-----+-----+

```

```

CREATE VIEW EmpProject(empNo, projNo, totalHours) AS
SELECT w.empNo, w.projNo, SUM(hoursWorked)
FROM Employee e, Project p, WorksOn w
WHERE e.empNo = w.empNo AND p.projNo = w.projNo
GROUP BY w.empNo, w.projNo;

```

```
SELECT * FROM EmpProject;
```

```
+-----+-----+-----+
| empNo | projNo | totalHours |
+-----+-----+-----+
| E0001 | P0007 | 10         |
| E0004 | P0001 | 40         |
| E0008 | P0001 | 30         |
| E0002 | P0008 | 15         |
| E0005 | P0002 | 40         |
| E0009 | P0002 | 35         |
| E0003 | P0006 | 20         |
| E0006 | P0003 | 25         |
| E0010 | P0003 | 20         |
| E0007 | P0004 | 38         |
+-----+-----+-----+
```

```
SELECT projNo FROM EmpProject WHERE projNo = 'P0007';
```

```
+-----+
| projNo |
+-----+
| P0007  |
+-----+
```

```
SELECT COUNT(projNo) FROM EmpProject WHERE empNo = 'E0001';
```

```
+-----+
| COUNT(projNo) |
+-----+
| 1             |
+-----+
```

```
SELECT empNo, totalHours FROM EmpProject GROUP BY empNo;
```

This provides an error, since there is no aggregate function (like SUM, AVG) being used with the GROUP BY function.

7.28

In order to maintain a materialized view, each operation would need to be considered.

If a new row is being inserted into the PART table, if partCost > 1000 and the partNo is not already in ExpensiveParts, the partNo would need to be inserted into the ExpensiveParts

view.

If a row is deleted, and the $\text{partCost} \leq 1000$, there would be no change needed

However if it had a $\text{partCost} > 1000$, there would need to be a check to determine if any other row with the same partNo has a $\text{partCost} > 1000$. If it happens to be so, partNo stays in the view, if not it would need to be deleted from the view.